Lecture 4

The Legendre transform and Hamilton's equations

Objectives. *Lecture 3 gave the first answer: a motion makes the action stationary, and its momentum is $p = \partial\mathcal{L}/\partial\dot{q}$. So far a state has been a position with a velocity. This lecture asks when the velocity can be traded for the momentum. The momentum is a different kind of object, a covector, and the trade is the Legendre transform; it works when the Lagrangian is hyperregular, and then Lagrange's n second-order equations become Hamilton's $2n$ first-order equations on the space of positions and momenta, the* phase space*, with an action principle of their own. This is the rung from the Lagrange layer to the Hamilton layer, on which the rest of the course stands.*

4.1 Momentum is a covector

Throughout, q ranges over an open set $U \subseteq \mathbb{R}^n$, a patch of generalised coordinates on a configuration space Q or $\mathbb{R}^n$ itself, and $\mathcal{L}(q, \dot{q}, t)$ is a C^2 Lagrangian with momentum $p = \partial\mathcal{L}/\partial\dot{q}$ (Definition 3.9). Momentum has n components, like the velocity. The two transform differently.

Let $\tilde{q} = \varphi(q)$ be other generalised coordinates, with φ a C^2 bijection onto an open set $\tilde{U}$ with invertible derivative (the transition map of Lemma 2.5). Velocities transform by $\dot{\tilde{q}} = D\varphi(q)\,\dot{q}$ (Equation (2.4)). Suppose the same system is described in the new coordinates by a Lagrangian $\tilde{\mathcal{L}}$ with the same value at the same state,

$$\mathcal{L}(q, \dot{q}, t) = \tilde{\mathcal{L}}\big(\varphi(q), D\varphi(q)\,\dot{q}, t\big), \tag{4.1}$$

as $T - V$ is, and as Theorem 3.6 builds $\tilde{\mathcal{L}}$ (read backwards). Write $\tilde{p} = \partial\tilde{\mathcal{L}}/\partial\dot{\tilde{q}}$.

Proposition 4.1 (The covector law)**.** *At corresponding states,*

$$p = D\varphi(q)^{\mathsf{T}}\,\tilde{p}, \qquad \text{that is} \qquad \tilde{p} = D\varphi(q)^{-\mathsf{T}}\,p, \tag{4.2}$$

and the pairing is invariant: $\tilde{p}\cdot\dot{\tilde{q}} = p\cdot\dot{q}$.

Proof. Differentiate (4.1) in $\dot{q}_j$. The kth velocity slot of $\tilde{\mathcal{L}}$ is $\sum_j(\partial\varphi_k/\partial q_j)\dot{q}_j$, so the chain rule gives $p_j = \sum_k \tilde{p}_k\,\partial\varphi_k/\partial q_j$, which is (4.2). Then $\tilde{p}\cdot D\varphi\,\dot{q} = (D\varphi^{\mathsf{T}}\tilde{p})\cdot\dot{q} = p\cdot\dot{q}$. ∎

Velocities are multiplied by $D\varphi$, momenta by $D\varphi^{-\mathsf{T}}$; the two agree only when $D\varphi$ is orthogonal. At a fixed configuration the velocities form a vector space $V \cong \mathbb{R}^n$, and a momentum p defines the linear function $v \mapsto p\cdot v$ on it: an element of the *dual space* V^*. When the components of vectors change by a matrix A, the components of dual vectors

change by $A^{-\mathsf{T}}$, so that the pairing is kept. Proposition 4.1 is this rule with $A = D\varphi(q)$: *a momentum is a covector*, a linear function of velocities, and p_j is its value on the velocity $\dot{q} = e_j$.

The picture

A covector $p \neq 0$ is drawn as the stack of parallel, equally spaced lines (hyperplanes, for $n > 2$) $\{v : p \cdot v = k\}$, $k \in \mathbb{Z}$. The number $p \cdot v$ is the signed number of lines that the arrow v, drawn from the origin, crosses, counting the last fraction. A large p has closely packed lines. A change of coordinates carries the arrow and the stack along together, so the count cannot change: that is the invariance of the pairing (Figure 4.1).

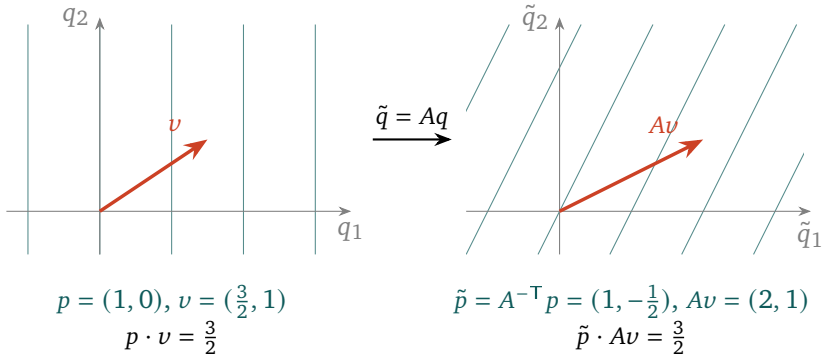


Figure 4.1: A covector as a stack of lines, crossed by a vector. The shear $A = \left(\begin{smallmatrix} 1 & 1/2 \\ 0 & 1 \end{smallmatrix}\right)$ tilts the lines $q_1 = k$ into $\tilde{q}_1 - \frac{1}{2}\tilde{q}_2 = k$ and moves the arrow; the arrow still crosses one and a half lines. The shear changes the first component of the vector and the second component of the covector.

Phase space

After the trade a state will be a configuration together with a covector: in a patch, a point $(q, p) \in U \times \mathbb{R}^n \subseteq \mathbb{R}^{2n}$, and on a change of patch

$$(q, p) \longmapsto \big(\varphi(q),\, D\varphi(q)^{-\mathsf{T}} p\big). \tag{4.3}$$

This is the *phase space*. In Lecture 1 phase space was the space of positions and velocities; from now on it is the space of positions and momenta. Over the whole of Q the set of such pairs is called the *cotangent bundle* T^*Q; we work in one patch.

4.2 The Legendre transform of a convex function

At fixed q and t the trade $\dot{q} \mapsto \partial\mathcal{L}/\partial\dot{q}$ is the gradient map of one function. When can a gradient map be inverted?

Theorem 4.2 (The Legendre transform). *Let $U \subseteq \mathbb{R}^n$ be open and convex, and let $f \in C^2(U)$ have a positive-definite Hessian $D^2 f$ at every point.*

1. *The* Legendre map $x \mapsto \nabla f(x)$ *is injective on U, its image U^* is open, and its inverse $p \mapsto x(p)$ is C^1. Define the* Legendre transform

$$g(p) = p \cdot x(p) - f\big(x(p)\big), \qquad p \in U^*. \tag{4.4}$$

2. $g \in C^2(U^*)$, $\nabla g(p) = x(p)$, *and* $D^2 g(p) = \big(D^2 f(x(p))\big)^{-1}$ *is positive definite.*
3. *The transform is* involutive*: the Legendre map of g is $p \mapsto x(p)$, a bijection of U^* onto U, and the transform of g is f.*
4. Young's inequality: $p \cdot x \le f(x) + g(p)$ *for all $x \in U$, $p \in U^*$, with equality if and only if $p = \nabla f(x)$. Hence $g(p) = \max_{x \in U}\big(p \cdot x - f(x)\big)$.*

Proof. (1) For $x \neq y$ in U the segment between them lies in U, so $(\nabla f(y) - \nabla f(x)) \cdot (y - x) = \int_0^1 (y-x)^{\mathsf{T}} D^2 f(x + s(y-x))(y-x)\,\mathrm{d}s > 0$. The derivative $D^2 f$ of ∇f is invertible, so by the inverse function theorem ∇f has C^1 local inverses on open sets; hence U^* is open and the global inverse is C^1.

(2) By the chain rule, $\nabla g = x + Dx^{\mathsf{T}}\big(p - \nabla f(x(p))\big) = x(p)$. So ∇g is C^1, and $D^2 g = Dx = (D^2 f)^{-1}$, the derivative of an inverse map.

(3) $\nabla g = x(\cdot)$ has inverse ∇f, and at $p = \nabla f(x)$, $x \cdot p - g(p) = f(x)$ by (4.4).

(4) Fix p, let $x_p = x(p)$, $h(x) = p \cdot x - f(x)$ and $u(s) = h(x_p + s(x - x_p))$ for $0 \le s \le 1$. Then $u'(0) = (p - \nabla f(x_p)) \cdot (x - x_p) = 0$ and $u'' < 0$ if $x \neq x_p$, so $h(x) = u(1) < u(0) = g(p)$ unless $x = x_p$. ∎

The pictures

For $f(x) = \cosh x - 1$, $g(p) = p\,\operatorname{arsinh} p - \sqrt{1+p^2} + 1$. By (4), $g(p)$ is the *largest vertical gap* between the line px and the graph of f; by (3), the graph is the *envelope* of the tangents $y = px - g(p)$ (Figure 4.2).

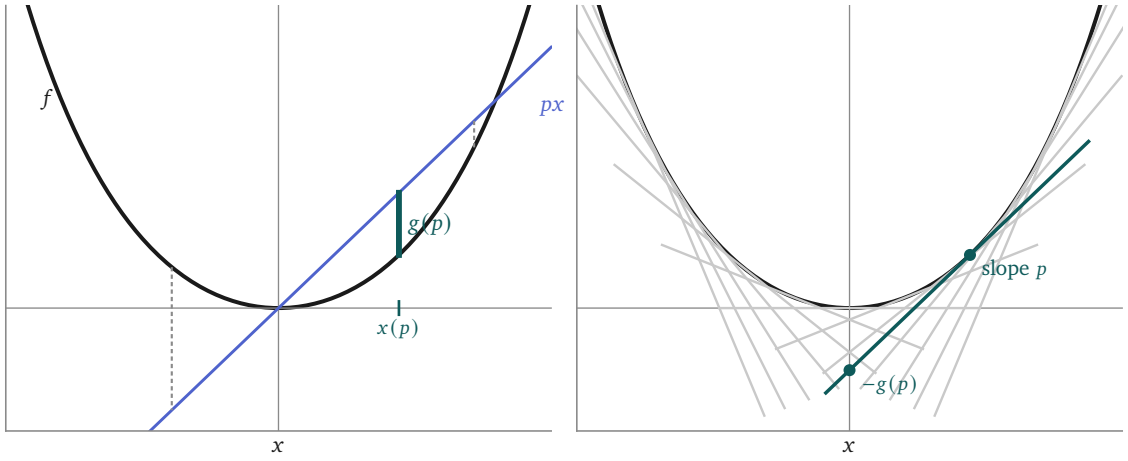


Figure 4.2: $f(x) = \cosh x - 1$. Left, the largest gap: for $p = 1.2$ the gap $px - f(x)$ is largest at $x(p)$, where it equals $g(p)$. Right, the envelope: the tangents $y = px - g(p)$, $p = -3, -2.5, \ldots, 3$.

Failure

Without a positive-definite Hessian the Legendre map need not be injective (Figure 4.3). For $f = x^3/3$ it folds at the inflection point: each $p > 0$ is x^2 at $x = \pm\sqrt{p}$, where $px - f(x) = \pm\frac{2}{3}p^{3/2}$. For the C^1 function $f = \frac{1}{2}\max(|x| - 1, 0)^2$ it is 0 on the whole flat piece $[-1, 1]$.

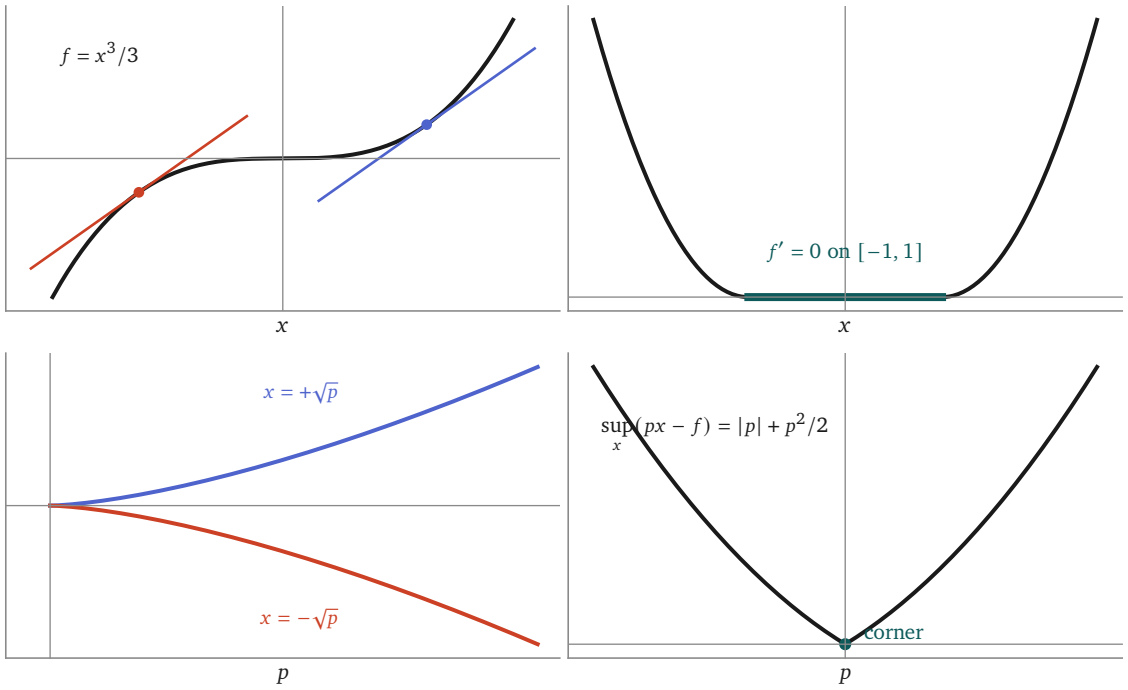


Figure 4.3: Left, an inflection point: two tangents of slope p (top), and two branches of $px - f(x)$ (bottom). Right, a flat piece (top): $\sup_x(px - f)$ has a corner (bottom).

4.3 The Legendre transform of a Lagrangian

As in Lecture 3, a Lagrangian is a C^2 function on $U \times \mathbb{R}^n \times I$, defined for all velocities.

Definition 4.3 (Fibre derivative; regular and hyperregular)**.** The *fibre derivative* of $\mathcal{L}$ is

$$\mathbb{F}\mathcal{L}(q, \dot{q}, t) = \left(q,\ \frac{\partial \mathcal{L}}{\partial \dot{q}}(q, \dot{q}, t),\ t\right),$$

and $W = \partial^2 \mathcal{L}/\partial \dot{q}\, \partial \dot{q}$ is the $n \times n$ Hessian in the velocities. $\mathcal{L}$ is *regular* if $\det W \neq 0$ everywhere, and *hyperregular* if it is regular and $\mathbb{F}\mathcal{L}$ is a bijection of $U \times \mathbb{R}^n \times I$ onto itself: at every (q, t) every momentum occurs exactly once. (Bijectivity alone is not enough: $\frac{1}{4}\dot{q}^4$ gives $p = \dot{q}^3$, whose inverse is not differentiable at 0.)

At fixed (q, t), $\mathbb{F}\mathcal{L}$ is the Legendre map of $\mathcal{L}(q, \cdot, t)$. Regularity means something for the motion: along a C^2 path the Euler–Lagrange equations (3.4) read

$$W\,\ddot{q} = \frac{\partial \mathcal{L}}{\partial q} - \frac{\partial^2 \mathcal{L}}{\partial \dot{q}\, \partial q}\,\dot{q} - \frac{\partial^2 \mathcal{L}}{\partial \dot{q}\, \partial t}. \tag{4.5}$$

If $\mathcal{L}$ is regular, this can be solved for $\ddot{q}$: a first-order system for the state $(q, \dot{q})$ of Lecture 1. Where W is singular, $\mathcal{L}$ does not determine the accelerations.

Regular is not enough for a global trade. $\mathcal{L} = e^{\dot{q}}$ has $W = e^{\dot{q}} > 0$ and $\mathbb{F}\mathcal{L}$ injective, but its momenta are only $p > 0$: the Hamiltonian $H = p \ln p - p$ below would live only on $p > 0$. Hyperregularity rules this out.

Lemma 4.4. *The derivative of $\mathbb{F}\mathcal{L}$ has determinant $\det W$. If $\mathcal{L}$ is hyperregular, the inverse of $\mathbb{F}\mathcal{L}$ is C^1.*

Proof. In the variables $(q, \dot q, t)$ the derivative is block lower triangular, with diagonal blocks I, W, 1. The rest is the inverse function theorem, as in Theorem 4.2(1). ∎

From here on, let $\mathcal{L}$ be hyperregular, and write the inverse as $(q, p, t) \mapsto (q, v(q, p, t), t)$: v is the velocity with $\partial\mathcal{L}/\partial\dot q(q, v, t) = p$. The *Hamiltonian* is

$$H(q, p, t) = p \cdot v(q, p, t) - \mathcal{L}\big(q, v(q, p, t), t\big), \qquad (q, p, t) \in U \times \mathbb{R}^n \times I. \tag{4.6}$$

At $p = \partial\mathcal{L}/\partial\dot q$ its value is $p\cdot\dot q - \mathcal{L} = E_{\mathcal{L}}$: *the Hamiltonian is the energy function of Theorem 3.14, written in momenta.*

Theorem 4.5 (Euler–Lagrange ⇔ Hamilton)**.** *Under these hypotheses:*

1. *$H \in C^2(U \times \mathbb{R}^n \times I)$, and at corresponding points $(q, p, t) = \mathbb{F}\mathcal{L}(q, v, t)$,*
$$\frac{\partial H}{\partial p} = v, \qquad \frac{\partial H}{\partial q} = -\frac{\partial \mathcal{L}}{\partial q}, \qquad \frac{\partial H}{\partial t} = -\frac{\partial \mathcal{L}}{\partial t}. \tag{4.7}$$
2. *A C^2 path $q(t)$ in U satisfies the Euler–Lagrange equations if and only if $(q(t), p(t))$, with $p = \partial\mathcal{L}/\partial\dot q(q, \dot q, t)$, satisfies* Hamilton's equations
$$\dot q = \frac{\partial H}{\partial p}, \qquad \dot p = -\frac{\partial H}{\partial q}. \tag{4.8}$$
3. *Every C^1 solution of (4.8) arises in this way.*

Proof. (1) v is C^1 by Lemma 4.4. Differentiate (4.6) by the chain rule:

$$\frac{\partial H}{\partial p_j} = v_j + \sum_k \Big(p_k - \frac{\partial \mathcal{L}}{\partial \dot q_k}\Big)\frac{\partial v_k}{\partial p_j}, \qquad \frac{\partial H}{\partial q_j} = \sum_k \Big(p_k - \frac{\partial \mathcal{L}}{\partial \dot q_k}\Big)\frac{\partial v_k}{\partial q_j} - \frac{\partial \mathcal{L}}{\partial q_j},$$

with $\mathcal{L}$ and its derivatives at (q, v, t). The brackets vanish by the definition of v; the same computation in t gives the third formula. The right sides of (4.7) are C^1, so H is C^2.

(2) Along the path, $(q, p, t) = \mathbb{F}\mathcal{L}(q, \dot q, t)$, so $v(q, p, t) = \dot q$, and the first equation of (4.8) holds by (1) whether or not q is a motion. By (1) again, $-\partial H/\partial q(q, p, t) = \partial\mathcal{L}/\partial q(q, \dot q, t)$, so the second equation reads $\dot p = \partial\mathcal{L}/\partial q$, which is (3.4).

(3) If (q, p) solves (4.8), then $\dot q = v(q, p, t)$ is C^1, so q is C^2, and $\mathbb{F}\mathcal{L}(q, \dot q, t) = (q, p, t)$: p is $\partial\mathcal{L}/\partial\dot q$ along the path, and (2) applies. ∎

In (4.7), $\partial H/\partial q$ holds p fixed and $\partial\mathcal{L}/\partial q$ holds $\dot q$ fixed.

A first-order flow

Write $z = (q, p)$ and let $J = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$, with $n \times n$ blocks. Hamilton's equations are

$$\dot z = J\,\nabla H(z, t), \tag{4.9}$$

with ∇ the gradient in all $2n$ variables: a first-order system with a C^1 right side, which has unique maximal solutions by Theorem 1.3 (cited), applied to (1.6). One function determines the whole vector field, a fixed matrix times a gradient; the rest of Part II studies what that implies. Along solutions,

$$\frac{\mathrm{d}H}{\mathrm{d}t} = \frac{\partial H}{\partial q}\cdot\dot q + \frac{\partial H}{\partial p}\cdot\dot p + \frac{\partial H}{\partial t} = \frac{\partial H}{\partial t} = -\frac{\partial \mathcal{L}}{\partial t},$$

by (4.8): Theorem 3.14 again. H is conserved when $\mathcal{L}$ does not contain t, and for the Lagrangians of Lecture 2 it is $T+V$ ((3.11)).

For the oscillator, $\mathcal{L}=\tfrac12 m\dot q^2-\tfrac12 m\nu^2 q^2$ gives $p=m\dot q$, $H=p^2/(2m)+\tfrac12 m\nu^2 q^2$, and Hamilton's equations $\dot q=p/m$, $\dot p=-m\nu^2 q$.

4.4 The phase-space action principle

Hamilton's equations have a variational principle of their own, in which q and p are varied independently. A *phase path* is a C^1 map $t\mapsto(q(t),p(t))$ on $[t_0,t_1]$ with $q(t)\in U$; its action is

$$S[q,p]=\int_{t_0}^{t_1}\Big(p\cdot\dot q-H(q,p,t)\Big)\,\mathrm{d}t. \tag{4.10}$$

It is *stationary with q fixed at the ends* if $\frac{\mathrm{d}}{\mathrm{d}s}S[q+s\eta,\,p+s\zeta]\big|_{s=0}=0$ for all C^1 maps $\eta,\zeta\colon[t_0,t_1]\to\mathbb{R}^n$ with $\eta(t_0)=\eta(t_1)=0$; nothing is asked of ζ at the ends. At $p=\partial\mathcal{L}/\partial\dot q$ the integrand is $p\cdot\dot q-H=\mathcal{L}$, by (4.6): this is the action of Lecture 3 with the momentum set free.

Theorem 4.6 (The phase-space action principle). *A phase path is stationary with q fixed at the ends if and only if it satisfies Hamilton's equations (4.8) on $[t_0,t_1]$.*

Proof. For small s the varied path keeps q in the open set U. Differentiate under the integral and integrate $p\cdot\dot\eta$ by parts (p is C^1):

$$\frac{\mathrm{d}}{\mathrm{d}s}S\Big|_{s=0}=\int_{t_0}^{t_1}\Big[\zeta\cdot\Big(\dot q-\frac{\partial H}{\partial p}\Big)-\eta\cdot\Big(\dot p+\frac{\partial H}{\partial q}\Big)\Big]\mathrm{d}t+\Big[p\cdot\eta\Big]_{t_0}^{t_1}. \tag{4.11}$$

The boundary term vanishes because η does at the ends. If Hamilton's equations hold, the integral vanishes. Conversely, take $\eta=0$ and any C^2 ζ vanishing at the ends: Lemma 3.3 gives $\dot q=\partial H/\partial p$. Then $\zeta=0$ and any such η give $\dot p=-\partial H/\partial q$. ∎

No ζ appears in the boundary term, because $\dot p$ does not appear in the integrand: p is free at the ends (Figure 4.4). (Asking also $\zeta=0$ at the ends changes nothing, since the proof used only such ζ.) Prescribing q at both ends is $2n$ conditions, as many as $2n$ first-order equations can generally meet; prescribing p as well would over-determine the motion.

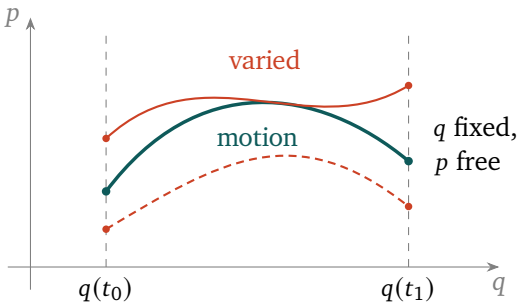


Figure 4.4: Varied paths in the phase-space action principle (Theorem 4.6): each starts on the line $q=q(t_0)$ and ends on the line $q=q(t_1)$, at any momentum. After Deriglazov, *Classical Mechanics*, Fig. 2.3.

4.5 Natural systems; Lagrange and Hamilton

Natural systems

Let $\mathcal{L} = \frac{1}{2}\dot{q}^{\mathsf{T}} M(q)\dot{q} - V(q)$ with $M(q)$ symmetric and positive definite (for Lecture 2's particles with fixed constraints, $M = \sum_i m_i\, Dr_i^{\mathsf{T}} Dr_i$, positive definite as Dr has rank n). Then $p = M(q)\dot{q}$ is a linear bijection in each fibre, $\mathcal{L}$ is hyperregular, and

$$H = \tfrac{1}{2}\, p^{\mathsf{T}} M(q)^{-1} p + V(q). \tag{4.12}$$

Lagrange and Hamilton

The two descriptions have the same motions wherever Theorem 4.5 applies. They are not equally good at everything; the entries marked with a lecture are paid off there.

	Lagrange (curves in Q)	**Hamilton** (flows on phase space)
Equations	n, second order	$2n$, first order, $\dot{z} = J\nabla H$
Needs	nothing to state them; W invertible to solve for $\ddot{q}$	a hyperregular $\mathcal{L}$ (here)
State	$(q, \dot{q})$; paths in Q may cross (Figure 1.1)	(q, p); for H without t, orbits never cross (Corollary 1.6)
Coordinates	any point transformation, and $\mathcal{L}$ is a scalar (Theorem 3.6)	point transformations lift by (4.3); also maps that mix q and p, such as $(q, p) \mapsto (p, -q)$ (Lecture 8)
Symmetry	Noether, from invariance of $\mathcal{L}$ (Theorem 3.13)	conserved $\Leftrightarrow$ zero bracket with H (Proposition 5.10)
Algebra	none	the Poisson bracket of observables (Definition 5.2)
Statistics	none	the flow preserves phase-space volume (Lecture 7)

Two gains are visible already: a state is a point, and for H without t orbits do not cross; and transformations may mix positions and momenta. One loss is visible too: the trade needs a hyperregular Lagrangian (Figure 4.3, and $e^{\dot{q}}$ above).

One misconception is worth naming. *"Only $\dot{p} = -\partial H/\partial q$ is dynamics."* From the Lagrangian side, $\dot{q} = \partial H/\partial p$ only restates the definition of p (the proof of Theorem 4.5(2)), and only the second equation is Euler–Lagrange. On phase space, where q and p are independent coordinates of a state, both are equations of motion: they come from one variational principle (Theorem 4.6).

Summary.

- *Handed on:* the phase space of pairs (q, p), with p a covector (Proposition 4.1, (4.3)); for a hyperregular $\mathcal{L}$, Hamilton's equations $\dot{z} = J\nabla H$ with $H = E_{\mathcal{L}}$ in momenta (Theorem 4.5, (4.9)); natural systems $H = \frac{1}{2}p^{\mathsf{T}}M^{-1}p + V$ ((4.12)); the phase-space action, with q fixed and p free at the ends (Theorem 4.6).
- *Next:* Lecture 5 asks what the observables of this phase space are and how they change: the Poisson bracket. Lecture 8 starts its time-dependent generating functions from the phase-space action.

References

Arnold, *Mathematical Methods of Classical Mechanics*, §§14–15 (pp. 61–65). Cannas da Silva, *Lectures on Symplectic Geometry*, Lecture 20 (pp. 121–124). Deriglazov, *Classical Mechanics*, §2.1 (pp. 91–101; the phase paths after his Fig. 2.3). Tong, *Lectures on Classical Dynamics*, §4.1 (pp. 82–88). Fortney, *A Visual Introduction to Differential Forms*, Figs. 1.5–1.8 (pp. 14–16), for the picture of a covector.